

Some Proofs

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Theorem 6.5.1

Let $b_k = \{\omega_k, v_k\}$ denote the Null basis pair that corresponds to $m \times 1$ vector valued function f_k . And let $\{b_1, \dots, b_n\}$ be a sequence of Null Basis pairs and $\{f_1, \dots, f_n\}$ be the corresponding $m \times 1$ vector valued functions, the following recursive updating formulae at step $k, k \leq n$, generates (A_k, B_k) , that are triangular input balanced.

We will prove this theorem by induction. So first, when $k = 1$, as initial condition, we have:

$$\begin{aligned} A_1 &= \omega_1 \\ L_1 &= [] \\ y_1 &= \frac{v_1}{\|v_1\|_2} \\ t_1 &= \sqrt{1 - |\omega_1|^2} \\ B_1 &= [\ t_1 y_1^* \] = \sqrt{1 - |\omega_1|^2} v_1^* \end{aligned}$$

Algorithm 1 Null Basis Realization

1. $L_k = [\ J_{k-1} L_{k-1} \quad t_{k-1} y_{k-1}^* \]$
 2. $t_k = \sqrt{1 - |\omega_k|^2}$
 3. $y_k = \frac{\prod_{i=1}^{k-1} \beta_{\omega_i, \frac{y_i}{\|y_i\|}} (\omega_k^*)^* v_k}{\|\prod_{i=1}^{k-1} \beta_{\omega_i, \frac{y_i}{\|y_i\|}} (\omega_k^*)^* v_k\|_2}$
 4. $J_k = I - (1 + \omega_k^*) y_k y_k^*$
 5. $A_k = \begin{bmatrix} A_{k-1} & 0 \\ t_k y_k^* L_k & \omega_k \end{bmatrix}, B_k = \begin{bmatrix} B_{k-1} \\ t_k y_k^* J_{k-1} J_{k-2} \cdots J_1 \end{bmatrix}$
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it's straight forward to see:

$$\begin{aligned}
A_1 A_1^* + B_1 B_1^* &= \omega_1 \omega_1^* + \sqrt{1 - |\omega_1|^2} v_1^* \left(\sqrt{1 - |\omega_1|^2} v_1^* \right)^* \\
&= |\omega_1|^2 + \sqrt{1 - |\omega_1|^2} v_1^* v_1 \sqrt{1 - |\omega_1|^2} \\
&= 1
\end{aligned}$$

then when $k = 2$,

$$\begin{aligned}
& A_2 A_2^* \\
&= \begin{bmatrix} A_1 & 0 \\ \sqrt{1 - |\omega_2|^2} y_2^* t_1 y_1 & \omega_2 \end{bmatrix} \begin{bmatrix} A_1^* & \left(\sqrt{1 - |\omega_2|^2} y_2^* t_1 y_1 \right)^* \\ 0 & \omega_2^* \end{bmatrix} \\
&= \begin{bmatrix} \omega_1 & 0 \\ \sqrt{1 - |\omega_2|^2} y_2^* \sqrt{1 - |\omega_1|^2} y_1 & \omega_2 \end{bmatrix} \begin{bmatrix} \omega_1^* & \sqrt{1 - |\omega_1|^2} y_1^* \sqrt{1 - |\omega_2|^2} y_2 \\ 0 & \omega_2^* \end{bmatrix} \\
&= \begin{bmatrix} \omega_1 \omega_1^* & \omega_1 \sqrt{1 - |\omega_1|^2} y_1^* \sqrt{1 - |\omega_2|^2} y_2 \\ \sqrt{1 - |\omega_2|^2} y_2^* \sqrt{1 - |\omega_1|^2} y_1 \omega_1^* & (1 - |\omega_1|^2) (1 - |\omega_2|^2) y_2^* y_1 y_1^* y_2 + \omega_2 \omega_2^* \end{bmatrix} \\
& B_2 B_2^* \\
&= \begin{bmatrix} B_1 \\ t_2 y_2^* J_1 \end{bmatrix} \begin{bmatrix} B_1 \\ t_2 y_2^* J_1 \end{bmatrix}^* \\
&= \begin{bmatrix} \sqrt{1 - |\omega_1|^2} v_1^* \\ \sqrt{1 - |\omega_2|^2} y_2^* (I - (1 + \omega_1^*) y_1 y_1^*) \end{bmatrix} \begin{bmatrix} \sqrt{1 - |\omega_1|^2} v_1^* \\ \sqrt{1 - |\omega_2|^2} y_2^* (I - (1 + \omega_1^*) y_1 y_1^*) \end{bmatrix}^* \\
&= \begin{bmatrix} (1 - |\omega_1|^2) v_1^* v_1 & \sqrt{1 - |\omega_1|^2} v_1^* \left(\sqrt{1 - |\omega_2|^2} y_2^* (I - (1 + \omega_1^*) y_1 y_1^*) \right) \\ \left(\sqrt{1 - |\omega_2|^2} y_2^* (I - (1 + \omega_1^*) y_1 y_1^*) \right) \left(\sqrt{1 - |\omega_1|^2} v_1^* \right)^* & \left(\sqrt{1 - |\omega_2|^2} y_2^* (I - (1 + \omega_1^*) y_1 y_1^*) \right) \left(\sqrt{1 - |\omega_2|^2} y_2^* (I - (1 + \omega_1^*) y_1 y_1^*) \right)^* \end{bmatrix}
\end{aligned}$$

GPT Proof

Then we can work on the proof:

Base case $k = 1$,

$$A_1 A_1^* + B_1 B_1^* = |\omega_1|^2 + t_1^2 y_1^* y_1 = 1$$

So the claim holds for $k = 1$.

Induction hypothesis

Assume for some $k \geq 2$ that (A_{k-1}, B_{k-1}) is input balanced:

$$A_{k-1} A_{k-1}^* + B_{k-1} B_{k-1}^* = I_{k-1}$$

Inductive step: show (A_k, B_k) is input balanced:

First, let's define

$$\begin{aligned}
a &= t_k y_k^* L_k \\
b &= t_k y_k^* P
\end{aligned}$$

Algorithm 2 GPT proof

1. Initialization ($k = 1$):

$$y_1 = \frac{v_1}{\|v_1\|_2}, A_1 = [\omega_1], t_1 = \sqrt{1 - |\omega_1|^2}, B_1 = t_1 y_1^*, L_1 = [], P_1 = I_m, J_1 = I_m - (1 + \omega_1^*) y_1 y_1^*$$

2. For $k = 2, \dots$, update L_k

$$L_k = \begin{bmatrix} J_{k-1} L_{k-1} & t_{k-1} y_{k-1} \end{bmatrix}$$

3. $t_k = \sqrt{1 - |\omega_k|^2}$.

4. Update product P_k :

$$P_k = J_{k-1} P_{k-1}$$

5. Compute the new orthonormal direction y_k ,

$$M_k = \prod_{i=1}^{k-1} \beta_{\omega_i y_i} (\omega_k^*)^*$$

and

$$y_k = \frac{M_k v_k}{\|M_k v_k\|_2}$$

6. Update J_k ,

$$J_k = I_m - (1 + \omega_k^*) y_k y_k^*.$$

7. Update (A_k, B_k) ,

$$A_k = \begin{pmatrix} A_{k-1} & 0 \\ t_k y_k^* L_k & \omega_k \end{pmatrix}, B_k = \begin{pmatrix} B_{k-1} \\ t_k y_k^* P_k \end{pmatrix}.$$

where $P = J_{k-1}J_{k-2} \cdots J_1$. Then,

$$A_k = \begin{pmatrix} A_{k-1} & 0 \\ a & \omega_k \end{pmatrix}, B_k = \begin{pmatrix} B_{k-1} \\ b \end{pmatrix}$$

Then we have,

$$A_k A_k^* = \begin{pmatrix} A_{k-1} A_{k-1}^* & A_{k-1} a^* \\ a A_{k-1}^* & a a^* + \omega_k \omega_k^* \end{pmatrix}, B_k B_k^* = \begin{pmatrix} B_{k-1} B_{k-1}^* & B_{k-1} b^* \\ b B_{k-1}^* & b b^* \end{pmatrix}$$

So

$$A_k A_k^* + B_k B_k^* = \begin{pmatrix} A_{k-1} A_{k-1}^* + B_{k-1} B_{k-1}^* & A_{k-1} a^* + B_{k-1} b^* \\ a A_{k-1}^* + b B_{k-1}^* & a a^* + b b^* + \omega_k \omega_k^* \end{pmatrix}$$

What remains to prove:

1. $A_{k-1} a^* + B_{k-1} b^* = 0$.
2. $a a^* + b b^* + \omega_k \omega_k^* = 1$.

For the second point, we need to prove:

$$a a^* + b b^* = t_k^2$$

But,

$$\begin{aligned} a a^* &= t_k y_k^* L_k L_k^* y_k t_k^* \\ b b^* &= t_k y_k^* P P^* y_k t_k^* \end{aligned}$$

So,

$$a a^* + b b^* = t_k y_k^* (L_k L_k^* + P P^*) y_k t_k^*$$

then we need to prove:

$$y_k^* (L_k L_k^* + P P^*) y_k = 1$$

similarly, we also need to show:

$$(A_{k-1} L_k^* + B_{k-1} P^*) y_k = 0$$